

CM0859 – MT5009 Type Theory

Course Introduction

Andrés Sicard-Ramírez

Universidad EAFIT

Semester 2025-2

Pacto pedagógico

Como miembros de la Universidad EAFIT, nos comprometemos a actuar de manera íntegra siguiendo los más altos estándares éticos y morales.

- Respeto
- Tolerancia
- Honradez
- Compromiso

Pacto pedagógico

Página web del curso

<https://asr.github.io/courses/cm0859-type-theory/2025-2/index.html>

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Conducto regular, fechas y porcentajes de las evaluaciones

La información está en la página web del curso.

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Responsabilidad compartida

- Profesor
- Estudiantes

Pacto pedagógico

Orientaciones para el curso

- Se recomienda seis horas de trabajo por semana (dos horas por cada hora de clase).
- Las clases son presenciales.
- La evaluación a la docencia es obligatoria.
- Se recomienda revisar periódicamente los canales de comunicación institucionales (EAFIT Interactiva y el correo institucional).
- El estudiante podrá entrar a un examen parcial a más tardar a los 20 minutos después de la hora asignada para su inicio.
- El estudiante podrá salir de un examen parcial después de transcurridos 30 minutos de la hora asignada para su inicio.

(Incomplete) Timeline

- Predicative theory of types Russell (1903)
- Theory of simple types Ramsey (1926)
- Untyped λ -calculus Church-Kleene-Rosser (1932–36)
- Natural deduction and sequent calculus Gentzen (1936)
- Cut-elimination and normalisation of proofs Gentzen (1936)
- Simple type theory Church (1940)
- Theory of categories Eilenberg-Mac Lane (1945)
- BHK interpretation of intuitionistic logic Brouwer-Heyting-Kolmogorov (1925, 1927)

(continued on next slide)

(Incomplete) Timeline

- Automath de Bruijn (1967)
- Typed domain logic Scott (1969–93)
- Formulae-as-types correspondence Curry-Howard (1969, 1980)
- Martin-Löf's type theories Martin-Löf (1973, 1979, 1986)
- Locally CCC and type theory Seely (1984)
- Calculus of constructions Coquand-Huet (1985)
- Homotopy type theory Awodey-Warren (2006)
- Univalence axiom Voevodsky (2006+)

Propositions-as-Types Principle

Three correspondence's levels

Wadler [Wad2015] introduces correspondence's levels by:

- (i) Propositions-as-types

For each proposition in the logic there is a corresponding type in the programming language—and vice versa.

Propositions-as-Types Principle

Three correspondence's levels

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(i) Propositions-as-types

For each proposition in the logic there is a corresponding type in the programming language—and vice versa.

(ii) Proofs-as-programs

For each proof of a given proposition, there is a program of the corresponding type—and vice versa.

Propositions-as-Types Principle

Three correspondence's levels

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(i) Propositions-as-types

For each proposition in the logic there is a corresponding type in the programming language—and vice versa.

(ii) Proofs-as-programs

For each proof of a given proposition, there is a program of the corresponding type—and vice versa.

(iii) Simplification of proofs as evaluation of programs

For each way to simplify a proof there is a corresponding way to evaluate a program—and vice versa.

Propositions-as-Types Principle

Other names

- The Curry-Howard correspondence/isomorphism

*[SU2006, p. viii].

Propositions-as-Types Principle

Other names

- The Curry-Howard correspondence/isomorphism
- The Brouwer - Heyting - Kolmogorov - Schönfinkel - Curry - Meredith - Kleene - Feys - Gödel - Läuchli - Kreisel - Tait - Lawvere - Howard - de Bruijn - Scott - Martin-Löf - Girard - Reynolds - Stenlund - Constable - Coquand - Huet - ... - correspondence*

*[SU2006, p. viii].

Natural Deduction (Conjunction and Implication)

Preliminaries

- Propositions: $A, B, C, \dots$
- Judgement: $A \text{ true}$ (assert, proposition A is true)
- Form of the inference rules

$$\frac{J_1 \quad \dots \quad J_n}{J} \text{ rule name}$$

where J (conclusion) and $J_1, \dots, J_n$ (premises) are all judgements.

- Types of inference rules: Introduction rules and elimination rules

Natural Deduction (Conjunction and Implication)

Inference rules for conjunction

- Introduction rule (composing information)

$$\frac{A \text{ true} \quad B \text{ true}}{A \wedge B \text{ true}} \wedge I$$

- Elimination rules (retrieving/using information)

$$\frac{A \wedge B \text{ true}}{A \text{ true}} \wedge E_1$$

$$\frac{A \wedge B \text{ true}}{B \text{ true}} \wedge E_2$$

Natural Deduction (Conjunction and Implication)

Inference rules for implication

- Introduction rule (hypothetical judgement)

$$\frac{\begin{array}{c} [A \text{ true}]^i \\ \vdots \\ B \text{ true} \end{array}}{A \supset B \text{ true}} \supset I^i$$

- Elimination rule (modus ponens)

$$\frac{A \supset B \text{ true} \quad A \text{ true}}{B \text{ true}} \supset E$$

Natural Deduction (Conjunction and Implication)

Example

A proof that $(A \wedge B) \supset (B \wedge A)$ true.

$$\frac{\frac{[A \wedge B \text{ true}]^i}{B \text{ true}} \wedge E_2 \quad \frac{[A \wedge B \text{ true}]^i}{A \text{ true}} \wedge E_1}{B \wedge A \text{ true}} \wedge I$$
$$\frac{B \wedge A \text{ true}}{(A \wedge B) \supset (B \wedge A) \text{ true}} \supset I^i$$

Typed Lambda Calculus (Product and Function Types)

- Types

$$\begin{array}{l} \sigma, \tau ::= \sigma \rightarrow \tau \\ \quad | \sigma \times \tau \end{array}$$

function type

product type

- λ -terms

$$\begin{array}{l} M, N ::= x \\ \quad | \lambda x. M \\ \quad | M N \\ \quad | \langle M, N \rangle \mid \text{fst } M \mid \text{snd } M \end{array}$$

variable

λ -abstraction

application

pairs and projections

- Judgement: $M : \sigma$ (λ -term M has type σ)

Typed Lambda Calculus (Product and Function Types)

Type assignment rules for product types

- Introduction rule (pair formation)

$$\frac{M : \sigma \quad N : \tau}{\langle M, N \rangle : \sigma \times \tau} \times I$$

- Elimination rules (pair projections)

$$\frac{M : \sigma \times \tau}{\text{fst } M : \sigma} \times E_1$$

$$\frac{M : \sigma \times \tau}{\text{snd } M : \tau} \times E_2$$

Typed Lambda Calculus (Product and Function Types)

Type assignment rules for function types

- Introduction rule (λ -abstraction)

$$\frac{\begin{array}{c} [x : \sigma]^i \\ \vdots \\ M : \tau \end{array}}{\lambda x. M : \sigma \rightarrow \tau} \rightarrow I^i$$

- Elimination rule (application)

$$\frac{M : \sigma \rightarrow \tau \quad N : \sigma}{M N : \tau} \rightarrow E$$

Typed Lambda Calculus (Product and Function Types)

Example

A proof that $\lambda h. \langle \text{snd } h, \text{fst } h \rangle : (\sigma \times \tau) \rightarrow (\tau \times \sigma)$.

$$\frac{\frac{\frac{[h : \sigma \times \tau]^i}{\text{snd } h : \tau} \times E_2 \quad \frac{[h : \sigma \times \tau]^i}{\text{fst } h : \sigma} \times E_1}{\langle \text{snd } h, \text{fst } h \rangle : \tau \times \sigma} \times I}{\lambda h. \langle \text{snd } h, \text{fst } h \rangle : (\sigma \times \tau) \rightarrow (\tau \times \sigma)} \rightarrow I^i$$

Correspondence's Levels

Example (propositions as types)

(implication) $A \supset B$ as $\sigma \rightarrow \tau$ (function type)

(conjunction) $A \wedge B$ as $\sigma \times \tau$ (product type)

Correspondence's Levels

Example (proofs as programs)

$$\frac{\frac{[A \wedge B \text{ true}]^i}{B \text{ true}} \wedge E_2 \quad \frac{[A \wedge B \text{ true}]^i}{A \text{ true}} \wedge E_1}{B \wedge A \text{ true}} \wedge I$$
$$\frac{B \wedge A \text{ true}}{(A \wedge B) \supset (B \wedge A) \text{ true}} \supset I^i$$

Proof

$$\frac{\frac{[h : \sigma \times \tau]^i}{\text{snd } h : \tau} \times E_2 \quad \frac{[h : \sigma \times \tau]^i}{\text{fst } h : \sigma} \times E_1}{\langle \text{snd } h, \text{fst } h \rangle : \tau \times \sigma} \times I$$
$$\frac{\langle \text{snd } h, \text{fst } h \rangle : \tau \times \sigma}{\lambda h. \langle \text{snd } h, \text{fst } h \rangle : (\sigma \times \tau) \rightarrow (\tau \times \sigma)} \rightarrow I^i$$

Program

References

- [SU2006] Morten-Heine Sørensen and Paul Urzyczyn. Lectures on the Curry-Howard Isomorphism. Vol. 149. Studies in Logic and the Foundations of Mathematics. Elsevier, 2006 (cit. on pp. 12, 13).
- [Wad2015] Philip Wadler. Propositions as Types. Communications of the ACM 58.12 (2015), pp. 75–84. DOI: [10.1145/2699407](https://doi.org/10.1145/2699407) (cit. on pp. 9–11).